

SMART INDIA HACKATHON 2025

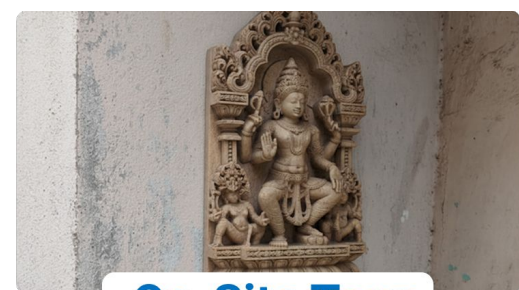


- Problem Statement ID - **SIH25052**
- Problem Statement Title - **AR-Based Cultural Heritage Preservation Platform**
- Theme - **Heritage & Culture**
- PS Category - **Software**
- Team ID - **72696**
- Team Name (Registered on portal) - **Invariants**



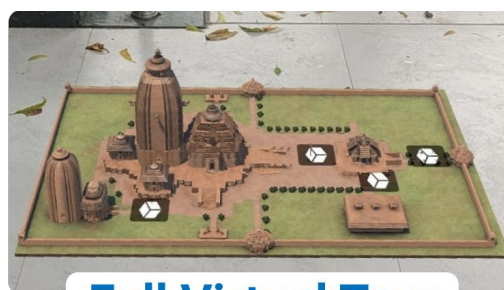
Proposed Solution

- An AR app with **3D reconstructed** and realistic compressed **GLB models** of heritage sites.
- **Dual-mode:** On-site AR overlays and immersive Virtual Tour.
- **Offline-first AR** for reliable use in rural areas, **packet split transfer** for **low-bandwidth** areas.
- **AI-driven, 20+ languages** personal guide for tailored **storytelling**.
- Local **economic boost** through **partnership**.
- **Crowd-sourced & moderated contributions** to enrich heritage records.
- Centralized **admin dashboard** for analytics, security, and content governance.



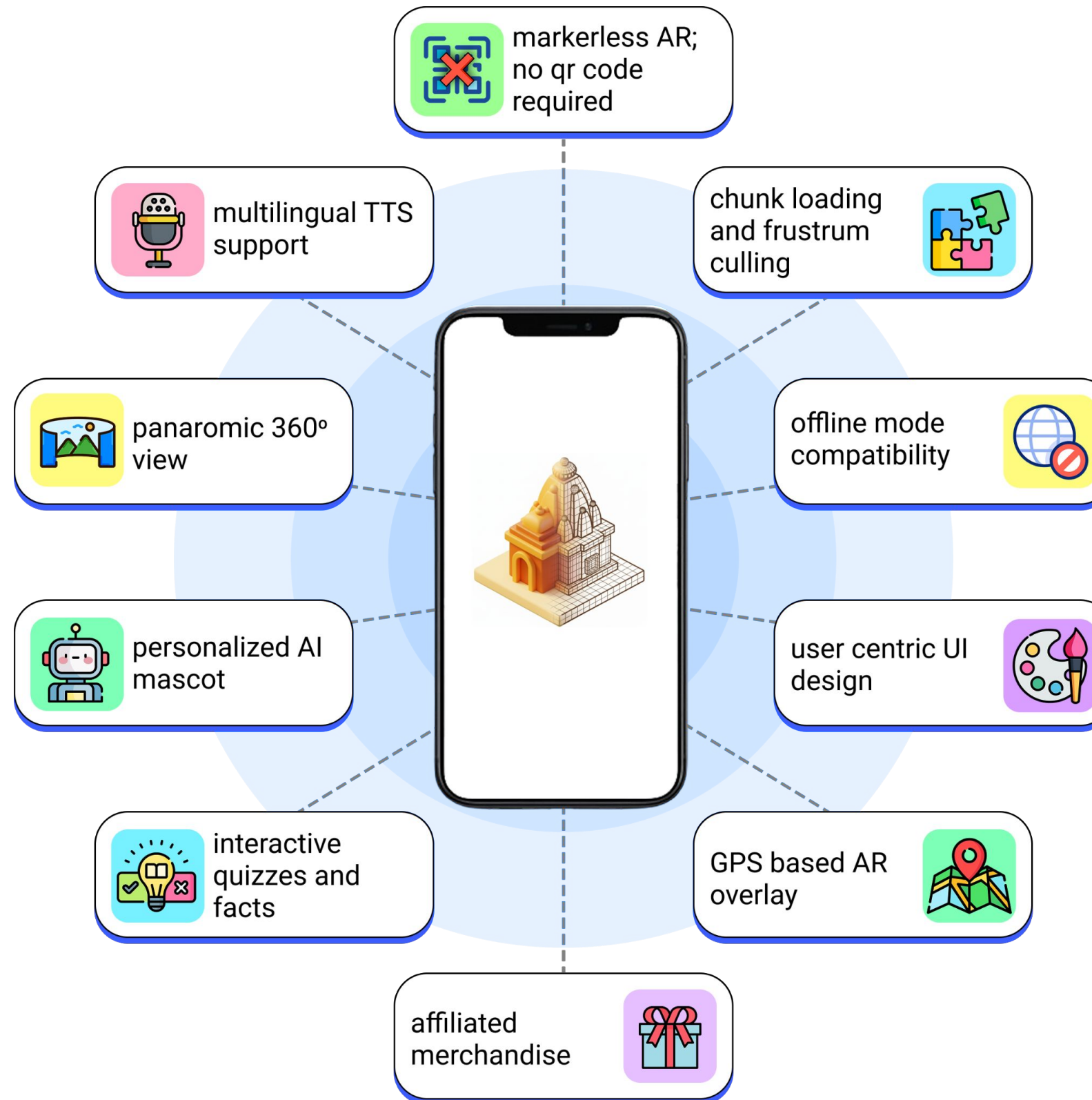
On-Site Tour

Experience history with immersive AR storytelling on location



Full Virtual Tour

lets you explore heritage sites from home with interactive 3d model



"bringing the past back to life forever"

It Addresses the Problem by

- **Digitally preserving heritage** via accurate **3D models** resilient to **decay**.
- **Scalable & reliable** with progressive streaming & offline-first design.
- Supports **sustainable** tourism, artisans livelihood, & cultural education.
- **Builds trust** with moderated contributions & **secure** data practices.



Android



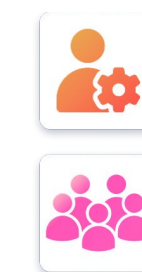
iOS



Meta Glasses

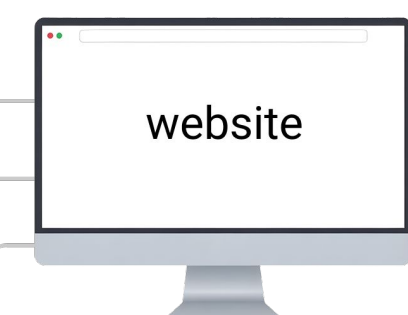


AR Headsets



Admin analytics and insights

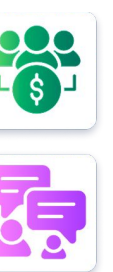
Crowd-sourced content moderation



website

Secure crowdfunding and donations

Feedback analysis and complaints handling



Admin analytics and insights

Crowd-sourced content moderation

TECHNICAL APPROACH



Tech Stack

App Frontend

Unity AR Foundation
Draco ARCore XRKit GLB Viewer
XR Interaction Kit

Website Frontend

Nextjs React
Typescript Tailwind CSS Axios
Zustand Shadcn UI oAuth Stripe

Backend (both)

Nodejs Expressjs
Javascript Python Django Prisma

APIs & AI

Google Maps Groq SDK
Gemini AI Clerk

Database / Cloud Storage

SQL
PostgreSQL Sheets Google Cloud

Deploy & Dev

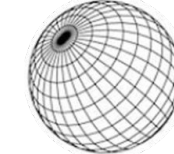
Vercel Render Git
Github Docker VS Code npm
Photoshop Blender Zbrush

Modular, scalable tech stack adaptable to future frameworks

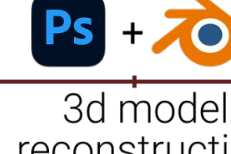


Our Working Prototype: [click here](#)

progress made so far: 60%



19mb
complex 3d models



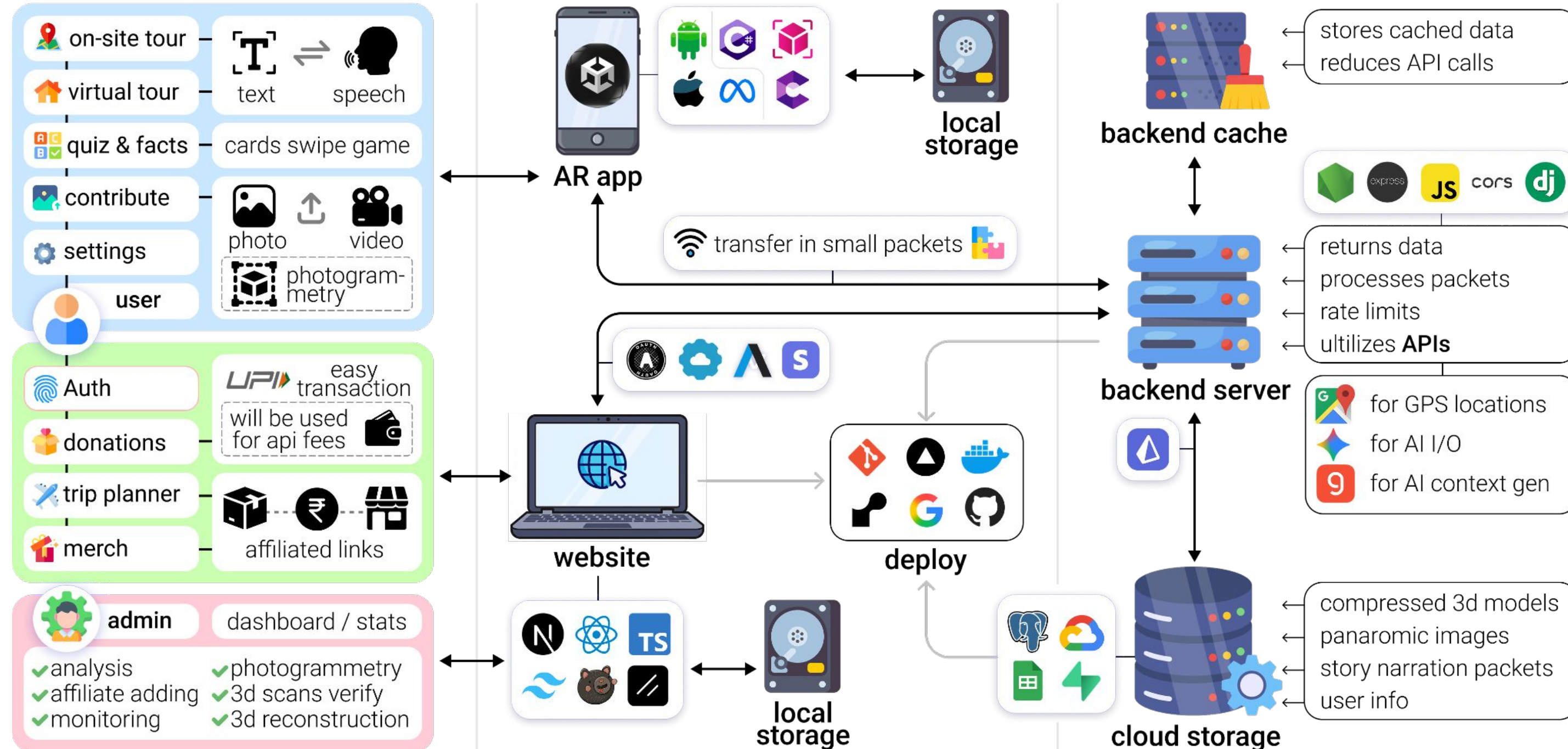
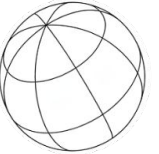
3d model reconstruction



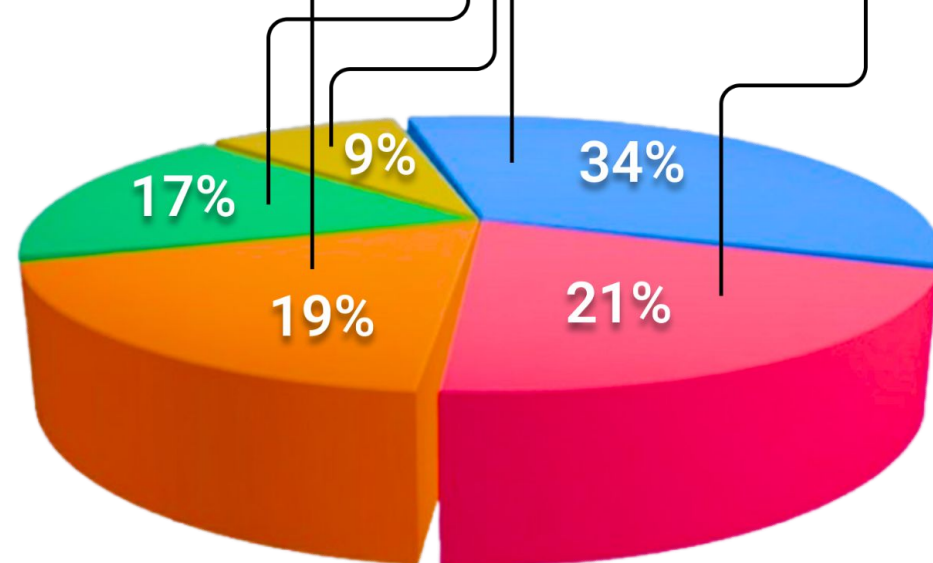
unity draco compression

7mb

compressed 3d models



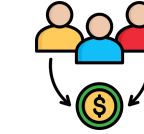
Financials

**Advertisements**
In-app ad revenue**Donations**
Voluntary user contributions**In-app Purchases**
Premium tours**Affiliated Links**
Earnings from partner links**Others** ₹32K - ₹38K
Data Privacy | Encryption |
Monitoring | Compliance**AR App** ₹65K - ₹72K
UI/UX Design | AR SDKs |
Testing | Accessibility**Backend** ₹70K - ₹80K
APIs/DB | Authentication |
Analytics | Integration**3D Models** ₹1.3L - ₹1.4L
Modeling | Acquisition |
Optimization | Management**Cloud** ₹75K - ₹85K
Compute | Storage |
Bandwidth | Security

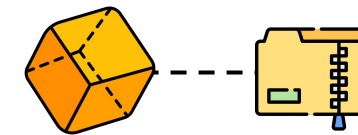
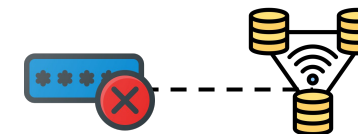
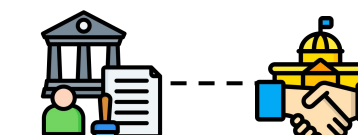
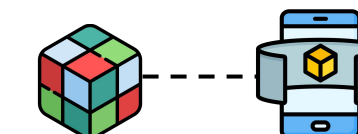
Feasibility

**Proven AR tech** built on **Unity, ARCore** and **ARKit** for **cross-platform** use.**Photogrammetry + AR** can digitize **1,000+ monuments** with high accuracy.**Offline-first AR** ensures access at **35%+ rural heritage sites** with weak connectivity.**Cloud hosting + 3D compression** reduce model size by **60%**, low-bandwidth access.

Viability

India's tourism economy to hit **\$250B by 2030**, **20% heritage-linked**.**Digital collectibles & gamified AR experiences** boost tourist spend by **20%**.AR tools in **schools & museums** reach **40M+ students**.**Public-private partnerships** with artisans enable affiliate sales, supporting local crafts.

Challenges and Mitigation

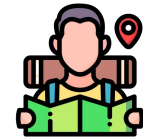
Heavy 3D files slow down devices**File compression** using **blender + unity's glb viewer** and **draco** compression & **LOD (Level of Detail)** cuts load time by up to **60%****Weak rural internet****Hybrid cache/CDN** may deliver content **2x faster** in low-connectivity zones**ASI/government scan approvals** needed**ASI partnerships + photogrammetry** accelerate **government approvals** for new AR contentDevice fragmentation (**Android/iOS**)Testing aims to cover **95% of major devices**; **fallback AR** planned for **older models**

Invariants

IMPACT AND BENEFITS



AR-enabled sites see a **15–20% rise in tourist engagement.**



82% of travelers prioritize cultural and heritage **AR experiences**



AR boosts knowledge **retention by 70%** vs traditional guides.



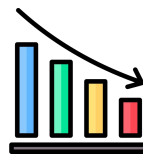
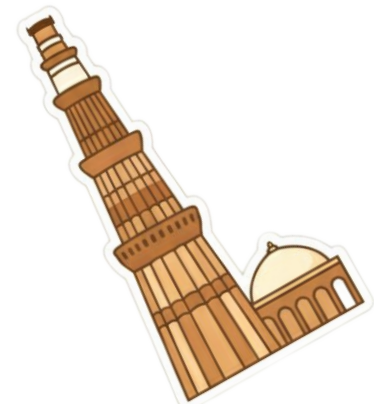
Storytelling in **20+ languages** can reach **650M+ tourists.**



Offline-first AR works in rural areas with **low connectivity**



India's AR/VR education market: **\$917M by 2025.**



Virtual tours can **cut physical strain** on sensitive sites by **15–20%.**



Heritage tourism in India projected at **\$57B by 2033.**

IMPACT



BENEFITS



Tourists: Multilingual AR tours solving info & language gaps.



Students: Interactive AR lessons boost engagement and recall by **70%.**



Local Guides: AR tools for storytelling, boosts **earnings by 30%.**



Artisans: Access a **\$2.5B+** digital marketplace, creating new income.



Government & ASI: AR preservation cuts costs **40%.**



Rural Communities: More visibility, **150K+** jobs supported.



Schools & Museums: Reach **40M+** learners with immersive edtech.



Global/NRI access, **amplifying** cultural reach.



Research Papers

AR in Cultural Heritage

- Augmented Reality in Cultural Heritage: An Overview of the Last Decade - Boboc et al., Applied Sciences, 2022 [\[Link\]](#)
- AI-driven 3D Restoration of Degraded Artefacts for Museum Digitalization - ScienceDirect, 2024 [\[Link\]](#)
- Deep Learning for Single Photo 3D Reconstruction of Cultural Heritage - ResearchGate, 2025 [\[Link\]](#)
- ItihasHub: A Digital Platform for the Preservation and Promotion of India's Cultural Heritage – Chaurasia et al., JETIR, 2025 [\[Link\]](#)

Government Applications

- National Digital Tourism Mission (NDTM) - Ministry of Tourism, Govt. of India, 2022 [\[Link\]](#)
- Preserving Indian Heritage with Digital Innovation - Kumara, Raghuvanshi & Surya, Int. Journal of Arts Architecture & Design, 2025 [\[Link\]](#)

User Demand & Community Insights

- “Best AR apps for history & museums” - r/augmentedreality, 2025 [\[Link\]](#)
- Visitors’ Acceptance of Wearable AR Technology in Museums - SAGE Journals, 2024 [\[Link\]](#)

Our Survey: [click here](#)

we conducted survey on Sept 2025
300+ respondents (Tourists,Students etc.)

Our Prototype: [click here](#)

Competitors Comparison	 SmarakaAR	 Augtraveler	 JuárezAR	 Xplore
Virtual Tour	✓	✓	✗	✓
Offline Mode	✓	✗	✓	✗
AI-Support	✓	✗	✗	✗
Multi-language	✓	✓	✗	✓
3D Reconstruction	✓	✗	✓	✓
Audio-Guide	✓	✓	✗	✓
Revenue Model	✓	✓	✗	✗

Industry Outlook

\$6.2B	AR/VR Tourism by 2030 (20% CAGR)
\$250B	India’s Tourism sector by 2030 40% heritage-linked
15–20%	Boost in engagement at AR-enabled heritage sites
70%	Higher cultural learning retention with AR
42	UNESCO Sites (India ranks 6th globally)

Primary Research(survey)

65.38%	of respondents reported a lack of information or guide support during heritage site visits.
50.00%	mentioned that high entry costs were a significant concern.
34.62%	pointed out language barriers in understanding the historical context.
30.77%	expressed that poor accessibility /connectivity made it difficult to reach or explore sites.

